Final Project - Dimensioning of divingboard

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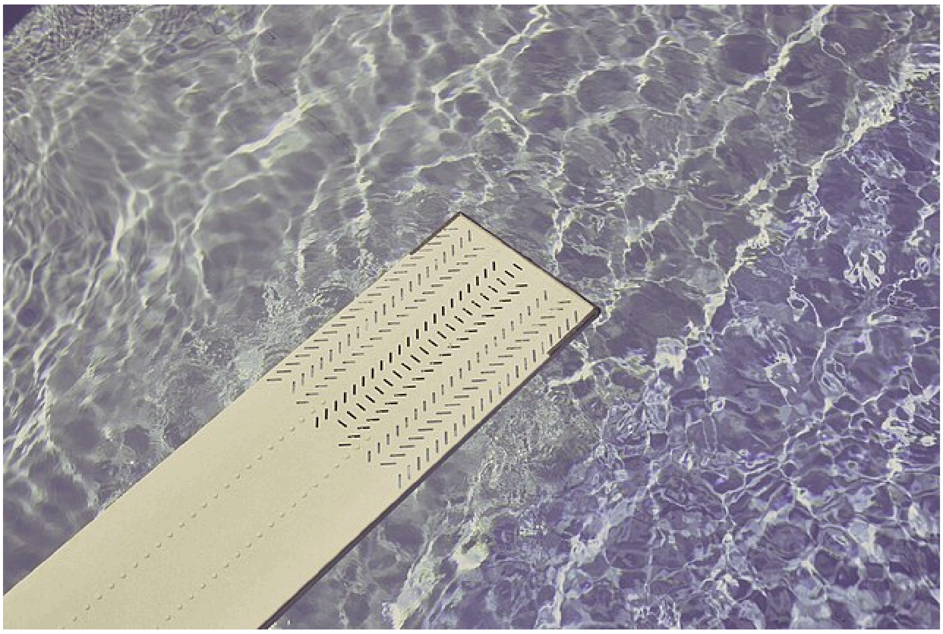

Final Project
FEM-modelling SE2860, 2026

Abstract

This report investigates the design optimization of diving boards with the goal of minimizing mass while satisfying deflection and strength constraints. Three loading scenarios are considered: (a) a person standing at the free end of a fixed diving board, (b) a person jumping from the edge, and (c) a board with one end simply supported and the other on a roller, representing a more realistic installation. The study compares homogeneous aluminum 6070-T6 with a wood-glass fibre epoxy composite, focusing on variations in board thickness and composite wood fraction. Both analytical beam theory and numerical finite-element analyses in COMSOL, including large-deformation effects, are used to evaluate maximum deflections and stresses. Results show that large-deformation analysis is essential for thin boards to avoid unrealistic deflection predictions. Optimization demonstrates that composite boards can achieve similar performance to aluminum with nearly 50% mass reduction, particularly in the simply supported configuration. The findings provide insight into material selection and geometric design strategies for lightweight, high-performance divingboards.

1. Problem background and definition

A divingboard is a common recreational structure that demonstrates elastic mechanics by bending under a person's weight, storing elastic energy that is released to propel the user upward. Boards are made from materials such as aluminum, laminated wood, and composite epoxies, which affect stiffness, durability, weight, and user experience.

This report aims to minimize the board's mass while meeting design constraints by varying material and thickness. Three cases are analyzed: (a) a fixed board with a person standing at the edge, (b) a person jumping at the edge, and (c) a board supported by a wall and a roller. Aluminum and composite materials are evaluated for each case.

2. Model definition

For Case (a) the diveboard is modelled after a fixed beam with a rigid connector at the start and a distributed load by the end of the board representing the user. For Case (b) the same model as Case (a) is used but instead of the distributed load a body is dropped 1 meter above the impact zone to study the physics. Lastly in Case (c) the diveboard is modeled with a roller constraint underneath to model how a diveboard usually is represented.

2.1 Geometry and materials

Figure 1 shows the geometry used in Case (a). The following assumptions are made; the divingboard is used by a person weighing 70kg, its end may not deflect more than 0.5 meter ($\Delta_{Allowed}$) or it will touch the water, one person uses the board at a time, it should not yield/break, and it is made of homogenous aluminum or a composite.

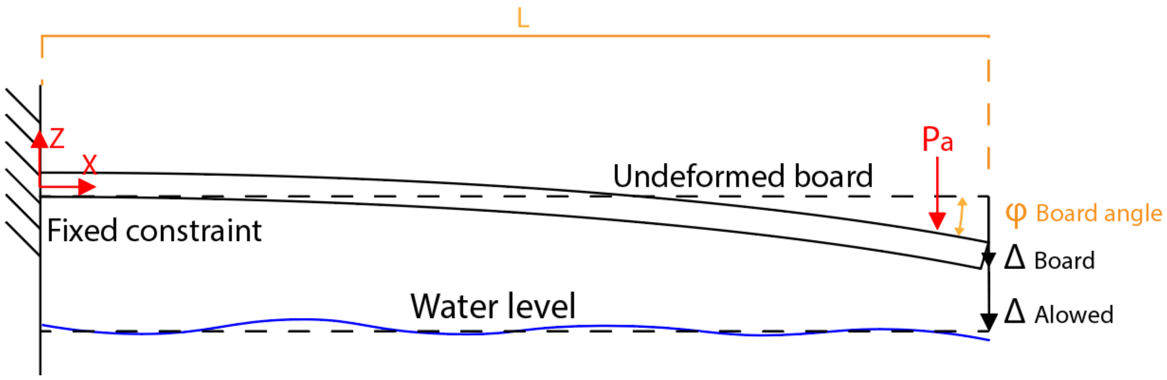

Figure 1: *Geometry of divingboard modeled as a fixed beam in the x-z-plane.*

2.2: Material parameters

The divingboard that will be used follows the 16 ft Maxiflex Model B Divingboard (MX-16), which uses aluminum 6070-T6 (Malmsten, n.d). The composites have fibres oriented along its length. In Case (a) both pine or plywood, while in the other cases, plywood and glassfiber epoxy is considered. Material and geometric parameters are provided in Appendix D and C respectively.

2.3: Relevant equations

Large deformation occurs when rotations are large and geometry changes significantly during loading, making small-deformation assumptions invalid. In COMSOL, this is handled by enabling geometric nonlinearities in the solver. Such analysis is required as strains and rotations exceed typical limits for small-deformation theory, ~5% strain or 10% rotation (Bonet & Wood, 2008). In finite elasticity, kinematics are defined in the deformed configuration using finite-strain measures derived from the deformation gradient (Bonet & Wood, 2008, Section 2.3), leading to Eqs. 1 and 2.

$$\mathbf{F} = \frac{\partial \mathbf{x}}{\partial \mathbf{X}} \tag{1}$$

Where $\boldsymbol{F}$ is the deformation gradient and $\boldsymbol{X}$ is the material, undeformed coordinates, and $\boldsymbol{x}$ are the deformed (current) coordinates. $\boldsymbol{F}$ captures stretch, rotation and shear of material points.

$$\mathbf{C} = \mathbf{F}^{T}\mathbf{F} \tag{2}$$

The right Cauchy-Green tensor, Eq (2), which is used to define the strain energy of the material, with $\boldsymbol{C}$ being frame independent.. The constitutive relationship is defined below Eq. 3.

$$\mathbf{S} = 2\frac{\partial \mathbf{W}}{\partial \mathbf{C}} \tag{3}$$

In COMSOL, finite elasticity uses the Second Piola-Kirchhoff stress $\boldsymbol{S}$, computed from a strain energy density function $\boldsymbol{W(C)}$, using hyperelastic finite-strain relations for the Second Piola–Kirchhoff stress, (Söderholm, 2008). $\boldsymbol{W}$ depends on the material model (e.g. Neo-Hookean). The equilibrium equation is expressed in the deformed configuration as in Eq. 4:

$$\boldsymbol{\nabla}\cdot\boldsymbol{\sigma} + \mathbf{b} = \mathbf{0} \tag{4}$$

Where $\boldsymbol{\sigma}$ is the cauchy stress, with $\boldsymbol{\nabla}\cdot\boldsymbol{\sigma}$ representing a vector of net internal forces per unit current volume and $\boldsymbol{b}$ being the body forces, for example gravity, using the local balance of linear momentum in the current configuration (Söderholm, 2008). COMSOL finds stresses by first calculating the deformation gradient, Eq. 3, the strain measures are then found using Green-Lagrange, Eq. 5.

$$\mathbf{E} = \frac{1}{2}(\mathbf{C} - \mathbf{I}) \tag{5}$$

Where $\boldsymbol{E}$ is the Green-Lagrange strain tensor and $\boldsymbol{I}$ is the identity matrix, from (Söderholm, 2008). A material model is selected, after Eq. 5 the 2nd Piola-Kirchoff stress is calculated (Eq. 3). Lastly the sought Cauchy stress is found using Eq. 6, (Bonet & Wood, 2008).

$$\sigma = \frac{1}{\mathbf{J}}\mathbf{F}\mathbf{S}\mathbf{F}^T \tag{6}$$

2.4: Boundary conditions.

2.4.1: Boundary conditions Case (a)

A fixed boundary condition is applied to the right hand side, see Figure 1, the user is modelled as a distributed load over w_F x l_F meter squared, see Figure 3. A gravity node is also applied to the divingboard.

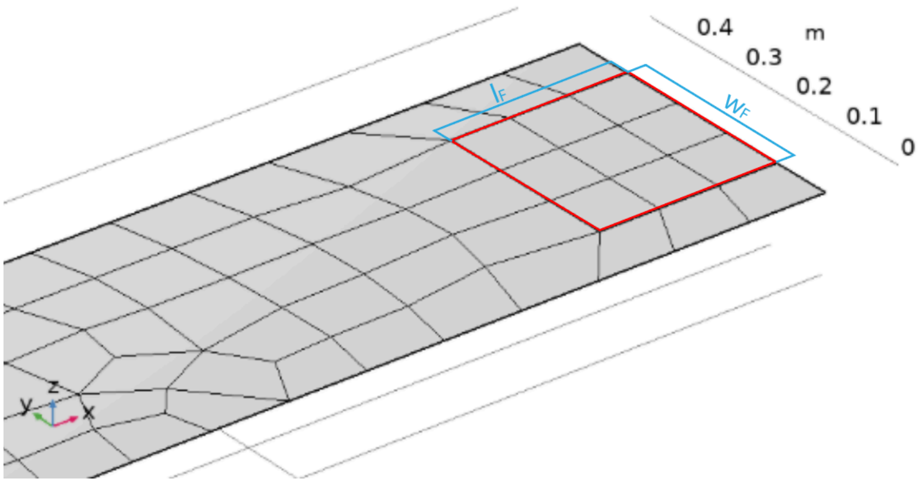

Figure 3: *User represented as a square with dimensions where* $w_F = l_F = 0.3$ *meter.*

2.4.2: Boundary conditions Case (b)

The distributed load applied on the free edge is replaced with a rigid connector which applies the same displacement experienced by the rigid body (the user), which is dropped from 1 meter. The drop was implemented as an initial velocity of the body $v_z = -\sqrt{2H_J g}$. Other boundary conditions remain the same. See Figure 4.

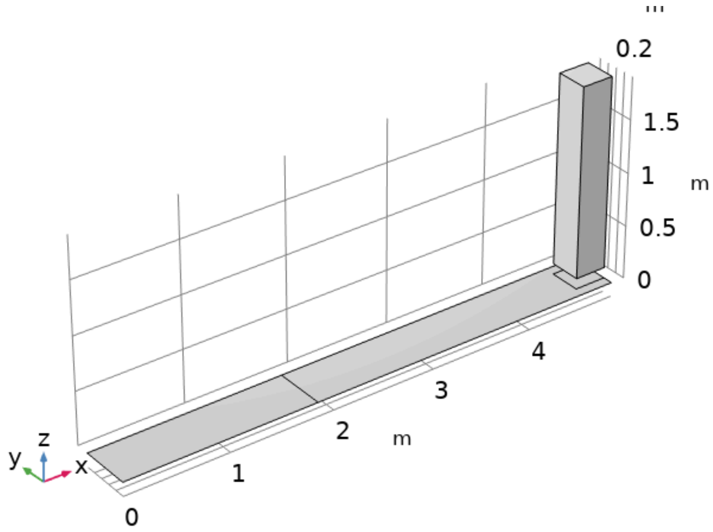

Figure 4: *Boundary conditions in Case (b), person modelled as box, fixed constraint on left boundary.*

2.4.3: Boundary conditions Case (c)

For Case (c) a roller constraint was added. The fixed constraint is replaced with a simple supported constraint. Constraints were modeled with prescribed displacement modules in COMSOL, see Figure 5.

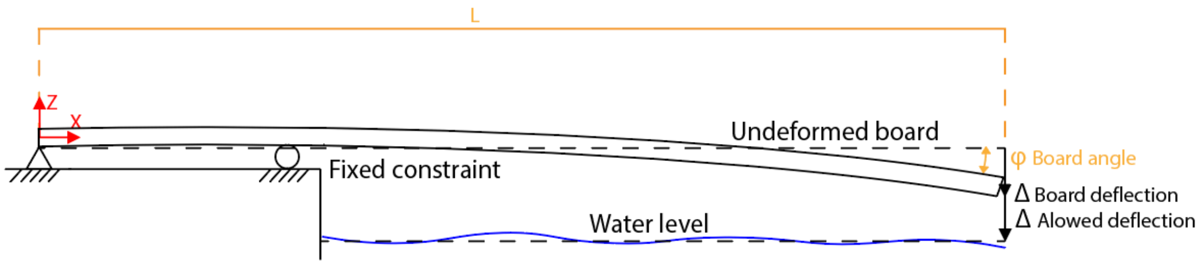

Figure 5: *Boundary conditions in Case (c)*

3. Numerical model

3.1: Mesh used for Case (a)

For Case (a), a numerical model is created in COMSOL using a layered shell (Figure 6). A free quadratic mesh with maximum element size 0.4 / *Nmesh* is used, where *Nmesh* is varied to ensure mesh convergence. Thickness, or thickness and wood fraction, are varied via optimization or parametric sweeps to meet the design criteria. For Case (a) *Nmesh* = 3; Case (b) and (c) *Nmesh* = 1 due to increasing computation times, while this did not have a significant effect of end deflection, as is presented in Figure 7.

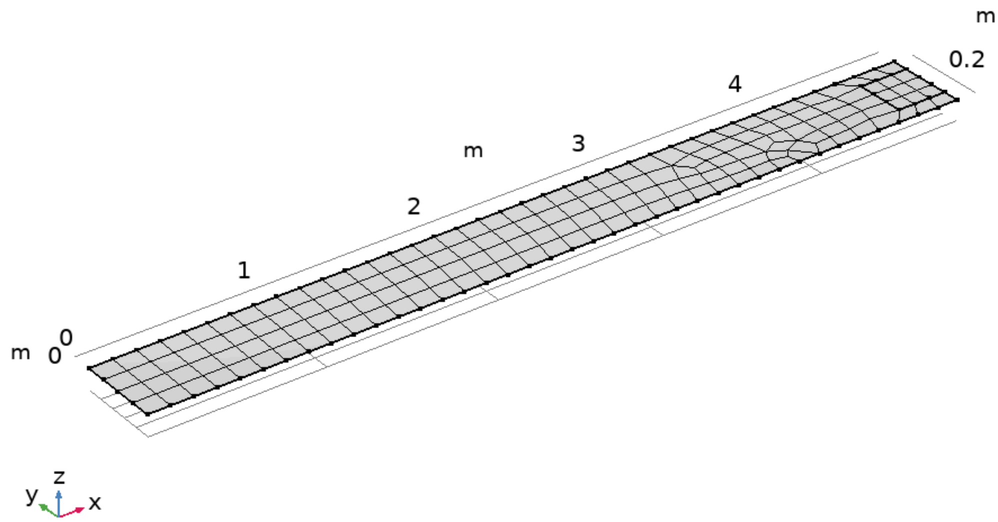

Figure 6: *Mesh used for Case (a). Free quadratic elements with maximum element size 0.4/Nmesh.*

3.2: Mesh convergence

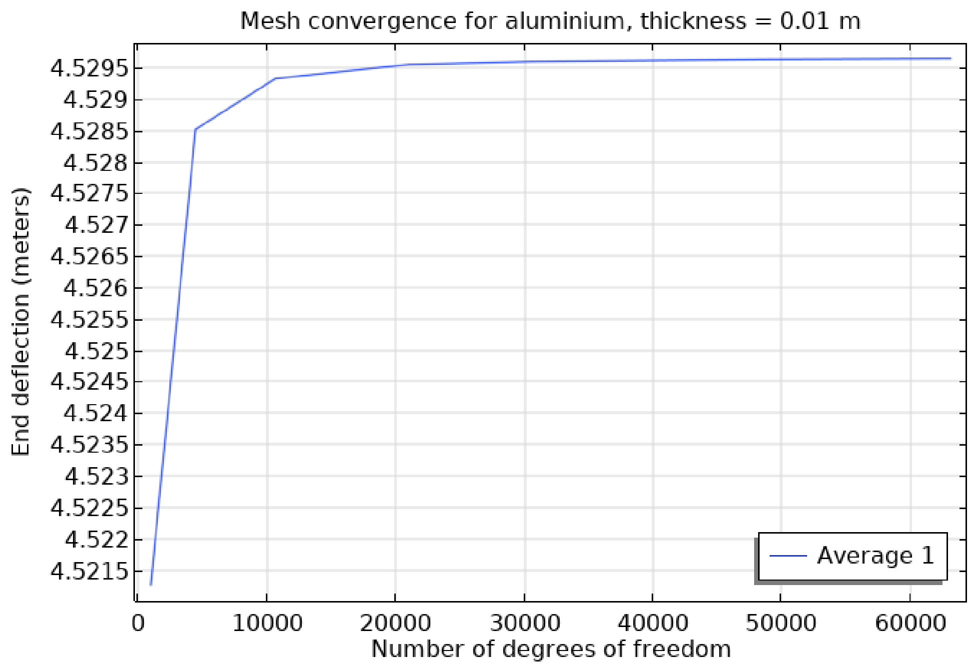

Figure 7: *Mesh convergence for Case (a), aluminium, thickness = 0.01 m.*

A mesh convergence study is conducted by varying the parameter *Nmesh*, which affects the number of degrees of freedom. The mesh does not change significantly after 4000 degrees of freedom (*Nmesh* = 3), while changing slightly between *Nmesh* = 1 and *Nmesh* = 2.

3.3 Additional solution steps

In Case (a), convergence was improved using load stepping, incrementally applying load and gravity from 0 to 100% using an auxiliary sweep. In Cases (b) and (c), a stationary study was first used to introduce gravity before the time-dependent analysis, reducing convergence issues and computation time.

4. Results and Discussion

4.1: Result for Case (a)

4.1.1: Analytical equations

The maximum deflection of the aluminum divingboard was analytically estimated using beam theory and elementary cases from (Alfredsson, 2014, Table 31.1), yielding expressions for deflection due to an edge point load (Eq. 7) and a distributed load (Eq. 8).

$$\Delta_{Board,pointload} = \frac{P_a L^3}{3E_A I} \tag{7}$$

$$\Delta_{Board,distributed\ load} = \frac{QL^3}{8E_A I} \tag{8}$$

Where I is the area moment of inertia for the cross section of the divingboard, $I = \frac{BH^3}{12}$ and Q is the gravity of the divingboard, $Q = gLBH$. Using superposition the total deflection becomes:

$$\Delta_{Board,Al} = \frac{4P_a L^3}{E_A BH^3} + \frac{3g\rho_A L^4}{2E_A H^2}. \tag{9}$$

To minimize the weight of the board while maintaining $\Delta_{Board} < \Delta_{Allowed}$, Δ_{Board} was plotted against thickness, in Figure 8.

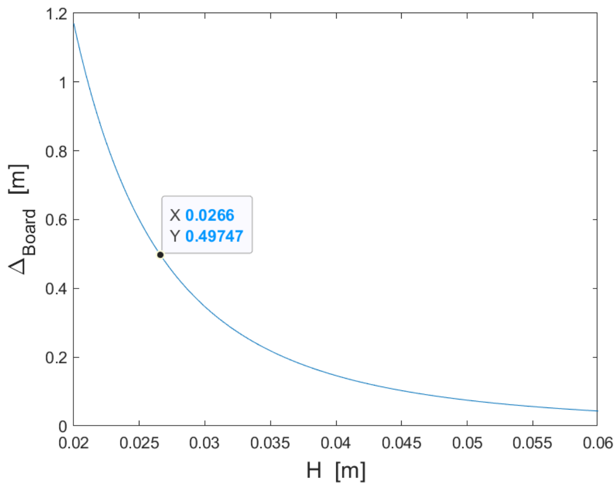

Figure 8: *Deflection of a homogenous divingboard with varying thickness*

Fig. 8 shows that the minimum allowed thickness analytically is 0.0266 meter, which gives weight 174.16 kg and Von Mises stress 54.234 MPa. The second constrain $\sigma_{s,A} > \sigma_e$ holds with good margin. To estimate deflection for the composite at varying thickness (H) and wood fraction (v_w), the top-surface stress was first computed, assuming the distribution shown in Figure 9.

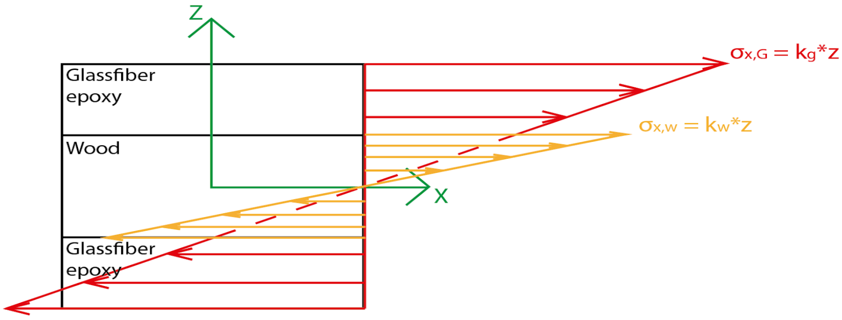

Figure 9: *Assumed stress distribution in the crosssection of the composite divingboard.*

Equilibrium of the divingboard gives the momentum Eq. 10.

$$M(x) = P(L - x) + \frac{mg(L-x)^2}{2L} \tag{10}$$

This together with the constraint at the intersection between pine and glass fibre epoxy $\varepsilon_{x,w}(v_w H/2) = \varepsilon_{x,G}(v_w H/2)$, where $\varepsilon_{x,w}(v_w H/2)$ and $\varepsilon_{x,G}(v_w H/2)$ is the strain of the wood and the glassfibre respectively at the intersection. This gives the system Eq. (11-12)

$$\frac{\sigma_{x,w}(v_w H/2)}{E_{x,Wood}} = \frac{\sigma_{x,G}(v_w H/2)}{E_{x,G}} \tag{11}$$

$$B \int_{-v_w H/2}^{v_w H/2} \sigma_{x,w}(z) z\, dz + 2 \int_{v_w H/2}^{H/2} \sigma_{x,G}(z) z\, dz = M(x) \tag{12}$$

From the equation system, the strain at the top of the divingboard can be found to be Eq. 13.

$$\varepsilon_{top} = C(H/2)M(x) \tag{13}$$

With

$$C = \frac{3}{2B((v_w H/2)^3 E_{x,P} + (3(v_w H/2)^2(1-v_w)H/2 + 3(v_w H/2)((1-v_w)H/2)^2 + ((1-v_w)H/2)^3)E_{x,G})}.$$

Now the displacement *w(x)* can be analyzed assuming

$$\frac{d^2 w}{dx^2} = \frac{\varepsilon_{top}}{H/2} = CM(x) \tag{14}$$

Solved using *w(0)=0* and $\frac{dw(0)}{dx} = 0$, gives

$$w(x) = C((P + \frac{mg}{2})\frac{Lx^2}{2} - (P + mg)\frac{x^3}{6} + \frac{mg}{2L}x^2) \quad (15)$$

Where the mass is

$$m = BLH(v_w\rho_{Wood} + (1 - v_w)\rho_G) \quad (16)$$

The maximum deflection can then be calculated at $x = L$ in Eq. (17).

$$\Delta_{Board,Composite} = C(\frac{PL^3}{3} + \frac{QL^3}{8}) \quad (17)$$

Equation (16) and (17) were put into MATLAB, yielding different results for pine and plywood since ρ_{Wood} and $E_{x,Wood}$ differ, see Appendix D. The parameters v_w and H were varied and the combination yielding the lowest mass while maintaining $\Delta_{Board,Composite} < \Delta_{Allowed}$ are displayed in Table 1.

Table 1: *Data for analytically optimized* v_w and H

Material	Mass [kg]	Thickness [m]	Proportion of wood [-]	Max equivalent stress [MPa]
Pine-epoxy composite	52.3915	0.0516	1.00	69.868
PlyWood-epoxy composite	94.6927	0.0486	0.910	68.071

4.1.2: Numerical solutions for comparison with analytical

To facilitate comparison with the analytical solution, a numerical solution for thickness = 0.01 meter is displayed in Table 2.

Table 2: *Numerical values for homogenous and composite for thickness = 0.05 meter*

Solution	Maximum end deflection [m]	Von mises stress [MPa]
Small deformations		
Analytical for aluminum	0.2066	54.234
Numerical for aluminium	0.19996	57.2
Analytical for Composite with Pine	0.3593	49.294
Numerical for Composite with Pine	0.34705	50.938
Analytical for Composite with Plywood	0.4459	62.658
Numerical for Composite with Plywood	0.4311	62.0769
Large Deformations		
Homogenous aluminium	0.19973	57.1
Composite with Pine	0.34574	50.752
Composite with Plywood	0.428669	61.84556

The composite featured in Table 2 is 90% wood and both cases have *thickness* = 0.05 meter.

The numerical solutions closely match the analytical results for all materials. For aluminum, the von Mises stress differs by 5.5%, with 54.23 MPa for the analytical solution and 57.2 MPa for the numerical solution, while the maximum end deflection differs by about 3.2% (0.2066 meter analytical versus 0.1996 meter numerical). For the pine composite, the numerical von Mises stress is 50.94 MPa compared to the analytical at 49.29 MPa, accompanied by a deflection difference of approximately 3.4% (0.359 meter analytical and 0.347 meter numerical). Similarly, for the plywood composite, the von Mises stress differs by less than 1%, with 62.66 MPa analytically and 62.08 MPa numerically, while the numerical deflection of 0.431 meter deviates by about 3.3% against analytical value of 0.446 meter.

4.1.3: Comparison between large and small deformations:

Large deformations may affect the simulations. A parametric sweep for the homogeneous aluminum case (Figure 10) shows that for small thicknesses, especially 0.01 meter, small-deformation analysis predicts an unphysical deflection of about 12 meter for a 5 meter board, while large-deformation analysis gives about 4.5 meter. The discrepancy decreases with increasing thickness. Therefore, since large-deformation analysis is more realistic for thin boards, geometric nonlinearities are enabled in COMSOL for all subsequent cases.

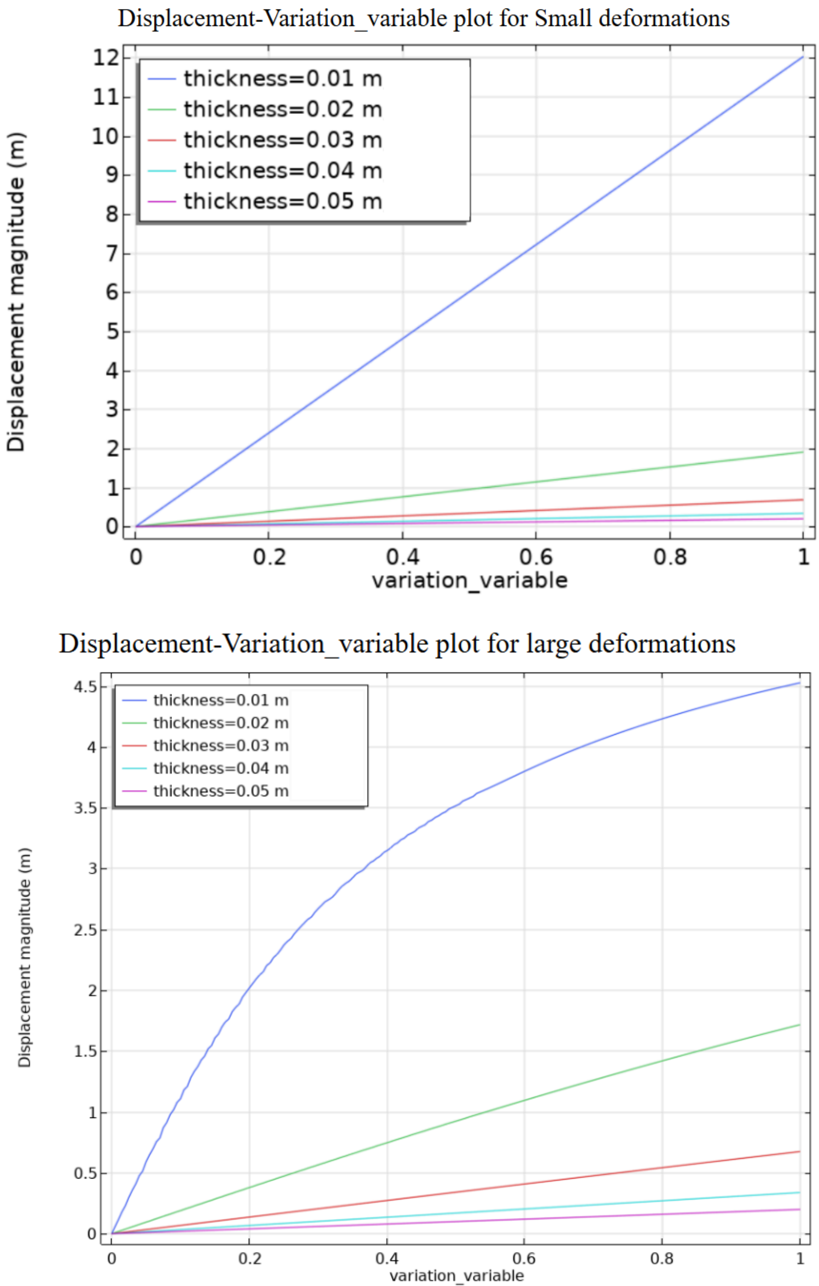

Figure 10: *Upper: Displacement-variation_variable plot for small deformations. Lower: Displacement-variation_variable plot for large deformations.*

4.1.4: Optimization

A key part of the task is to optimize the divingboard's mass so that the maximum displacement is under 0.5 meter, and that the maximum equivalent stress is lower than the yield stress. The max equivalent stress for all cases had yield stress way below of the yielding criteria. As mentioned previously the optimization uses the inbuilt optimization tool for Case (a). These values are featured in Table 3.

Table 3: *Numerical optimization of thickness and proportion of wood to satisfy a deflection under* $\Delta_{Allowed.}$

Material	Mass [kg]	Thickness [m]	Proportion of wood	Max equivalent stress [MPa]
Aluminium	226.05	0.034531	-	101.74
Pine-epoxy composite	51.624	0.0507	0.9989	71.619
PlyWood-epoxy composite	93.165	0.0476	0.9069	78.518

4.2: Results and discussion Case (b)

Although the built-in optimization could be used for Case (a), it was not practical for Cases (b) and (c) due to the time-dependent solver. The optimization only returned the final-time displacement, not the maximum over the time history, which often underestimated the maximum, therefore, Cases (b) and (c) were evaluated manually using parametric sweeps or single-length analyses.

4.2.1: Results and discussion of Case (b) - aluminum

The homogeneous aluminum divingboard was analysed using a rigid connector. The optimization was performed manually and is detailed in Appendix A. Figure 11 shows the displacement magnitude for the optimized thickness = 0.04585 meter.

Table 4: *Numerical optimization of thickness and proportion wood to satisfy a deflection under* $\Delta_{Allowed}$

Material	Mass [kg]	Thickness [m]	Proportion of wood
Aluminium	300.1467546	0.04585	-

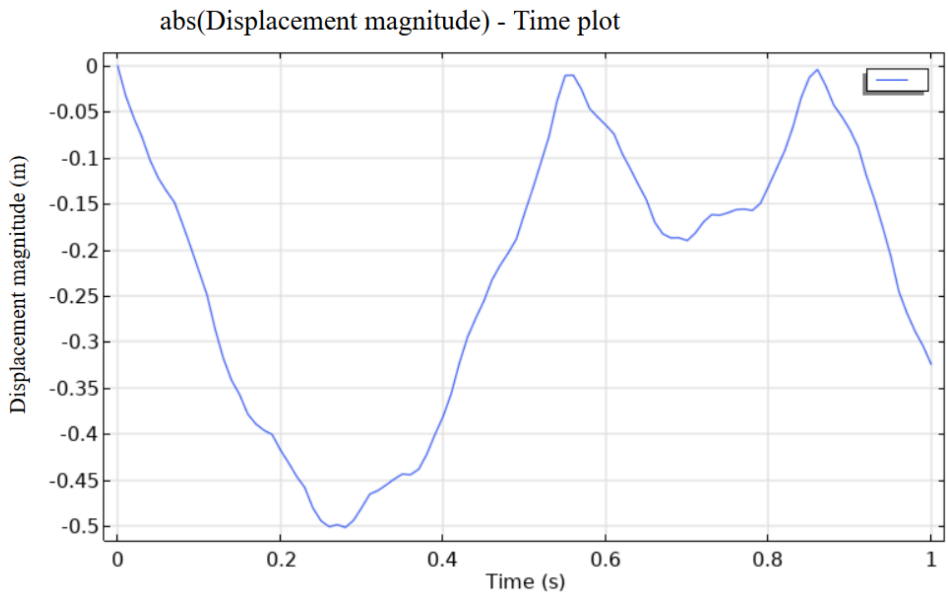

Figure 11: *Displacement magnitude-Time plot for homogenous aluminum with thickness 0.04585 meter.*

4.2.2: Comparison with result in Case (a)

The required thickness increases from 0.03453 to 0.04585 meter, a relative increase of 32.78%, which is reasonable since the divingboard must now account for the person's inertia. Accordingly, the mass increases proportionally from 226.05 to 300.15, also by 32.78%.

4.2.3: Results and discussion of Case (b) - composite

For the composite case, only plywood was considered to reduce computation time, and H was not optimized simultaneously with v_w. Instead, v_w = 0.9069 was fixed, based on optimized values from Case (a). This value is assumed to be load-independent, since neither bending stiffness nor weight depends on the loading case. A time-dependent study was then conducted in COMSOL following the setup in Section 2.4.2, with varying (*H*) (Figure 12). Curves below the "Allowed" limit do not satisfy the maximum deflection criterion.

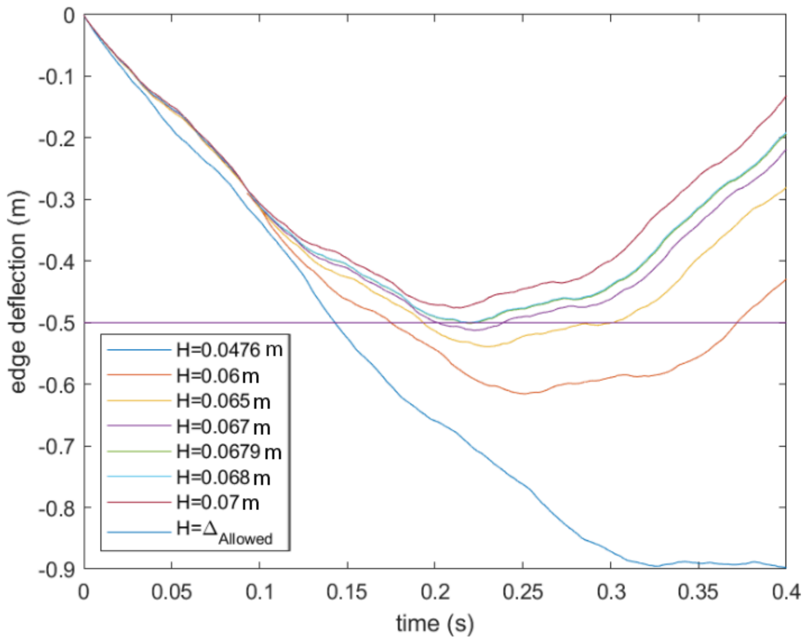

Figure 12: *Deflection of the edge of the divingboard plotted against time for different H.*

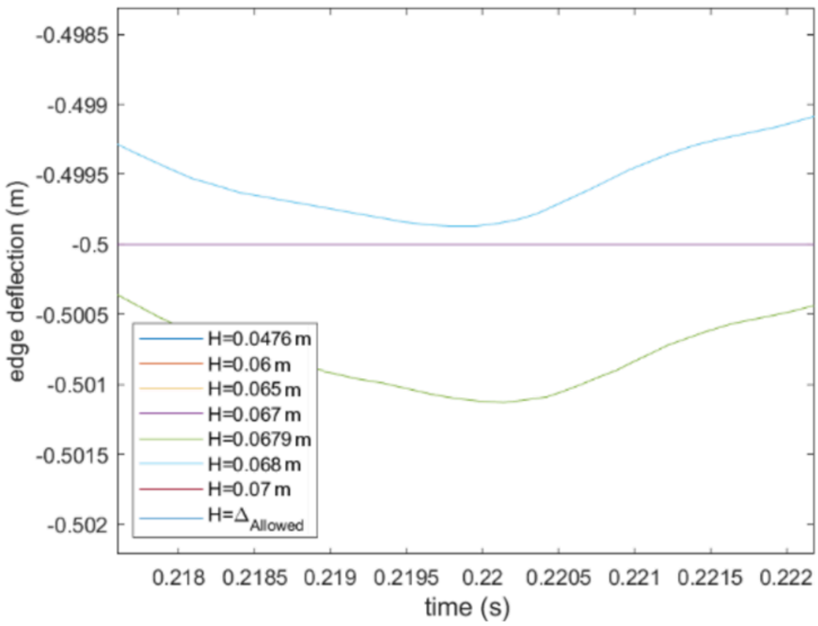

Figure 13: *Zoomed in version of Figure 12.*

Figure 13 shows that the minimum allowable thickness lies between ($H = 0.0679$) meter and ($H = 0.068$) meter. With the plywood fraction kept constant, this gives a minimum weight between 132.88 kg and 133.07 kg using Eq. 16. The selected thickness must also ensure that the maximum von Mises stress remains below the yield stress of both plywood and glass-fibre epoxy (Table D1, Appendix D). The maximum stress is assumed to occur at the time of maximum deflection, around 0.2 s (Figure 13). Figure 14 shows a peak von Mises stress of 72.642 MPa in the glass-fibre epoxy and significantly lower stresses in the plywood.

The condition ($\sigma_{s,G}$>72.642 MPa) MPa is therefore satisfied, and the plywood stress remains well below $\sigma_{s,Ply}$ with a good safety margin.

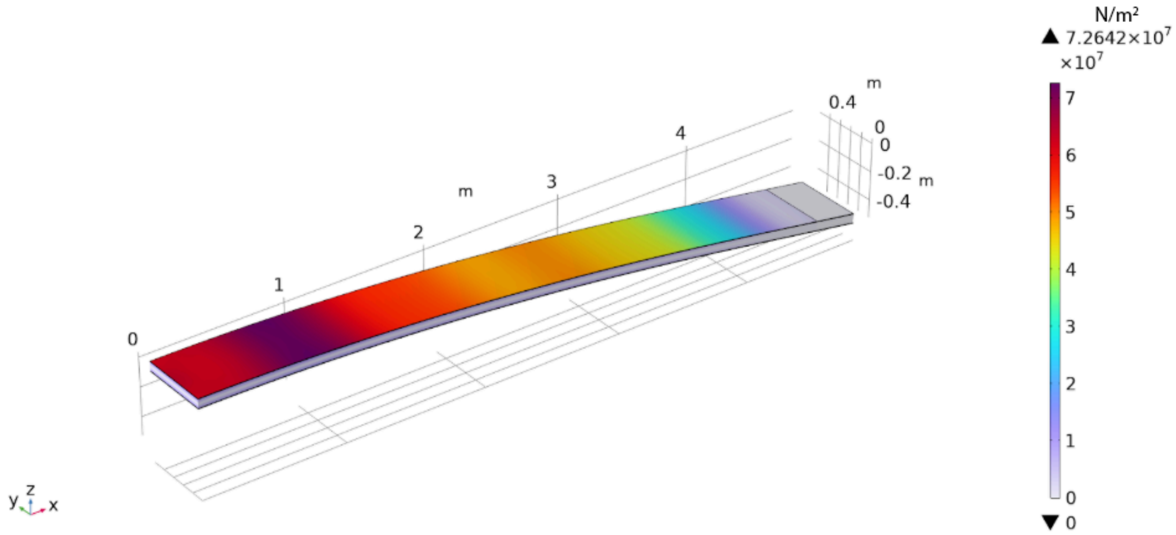

Figure 14: *Von Mises stress in divingboard 0.2 s after impact for H=0.0679 meter.*

4.3: Results and discussion of Case (c):

The full manual optimization procedure is featured in Appendix B.

Table 5: *Numerical optimization of thickness with* v_W*=0.9*

Material	Mass [kg]	Thickness [m]	Proportion of wood
Aluminium	199.661418	0.0305	-
PlyWood-epoxy composite	94.3271406	0.0482	0.9069

Using the optimized thickness from Case (b) of 0.068 meter resulted in an end displacement of 0.28 meter. The simulations show that the usable board length depends on thickness and roller placement. Replacing solid aluminum with a plywood–epoxy composite reduces the mass from 199.66 to 94.33 kg, a 49.25% decrease.

5. Conclusions

5.1: Case (a)

- Analytical beam theory gives a minimum aluminum thickness of 0.0266 meter (mass 174.2 kg), keeping deflection ≤ 0.5 meter and stress ~54 MPa.
- Large-deformation FEM shows that small-deformation theory overestimates thin-board deflections: 0.01 meter thickness predicts ~12 meter vs. ~4.5 meter in large-deformation analysis.
- For composite boards, analytical and numerical optimization shows that a wood–glass fibre epoxy configuration with a wood volume fraction of approximately $v_n \sim 0.9$ can satisfy the deflection constraint with substantially reduced mass compared to aluminum.
- Numerical simulations confirm that both aluminum and composite configurations remain well below their respective yield stresses, indicating that deflection is the governing design constraint in this case.

5.2: Case (b)

- Dynamic effects significantly increase the required structural stiffness compared to Case (a), necessitating a greater board thickness to satisfy the deflection limit.
- For homogeneous aluminum, the optimized thickness increases from 0.0345 meter (Case (a)) to 0.04585 meter, resulting in a mass increase from 226.1 kg to 300.1 kg, corresponding to a 32.8% increase due to inertial effects.
- For the composite configuration (plywood-glass fibre epoxy), the wood volume fraction was fixed at $v_n = 0.9069$, based on Case (a) optimization. The minimum allowable thickness was found to be 0.0680 meter, yielding a total mass of approximately 133 kg.
- Maximum von Mises stress occurs at the time of peak deflection (~ 0.2 **s**) and reaches ~ 72.6 MPa in the glass-fibre epoxy layer, while stresses in the plywood layer are considerably lower.

5.3: Case (c)

- The supported boundary condition with a roller substantially reduces the structural demands compared to the fixed-end configurations in Cases (a) and (b).
- For aluminum, an optimized thickness of 0.0305 meter results in a total mass of 199.7 kg, satisfying the maximum deflection criterion.
- For the plywood-glass fibre epoxy composite with $v_n = 0.9069$, the optimized thickness is 0.0482 meter, corresponding to a mass of 94.3 kg.
- This represents a 49.3% reduction in mass compared to the aluminum, while maintaining acceptable deflection levels (maximum end displacement < 0.5 meter).

6. AI

ChatGPT was used to summarize, shorten and spell check paragraphs and cite sources in the reference section. ChatGPT was also used to find certain sources and some information used in the theory section, these sources were checked thoroughly.

7. References

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8: Appendix A:

First a large span parametric sweep was conducted, as is seen in Figure A1

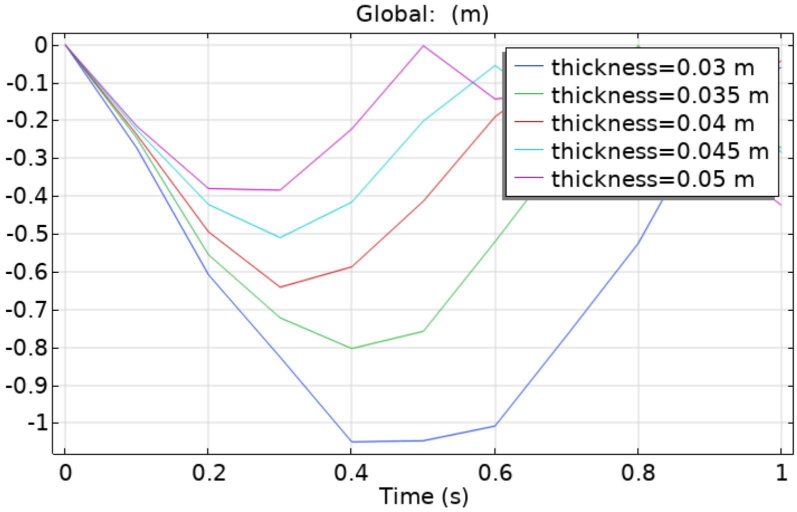

Figure A1: *Large span of parametric sweep.*

It can be seen from Figure A1 that the thickness which gives a deflection of around 0.5 meter is around 0.45, also note that the output times for the above were very crude, however the purpose was only to find a smaller scope. Values were then swept from 0.04 to 0.05, as is presented in Figure A2:

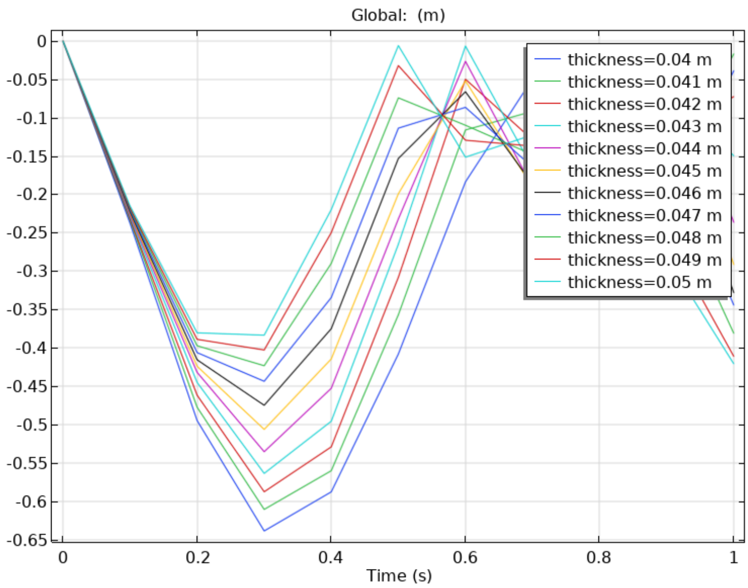

Figure A2: *Parametric sweep of deflection-time for thickness of 0.04 to 0.05 meter.*

From the figure above it was shown that the values closest to 0.5 end deflection were around 0.044 to 0.047, an additional parametric sweep was conducted for values in this range, as is seen in Figure A3.

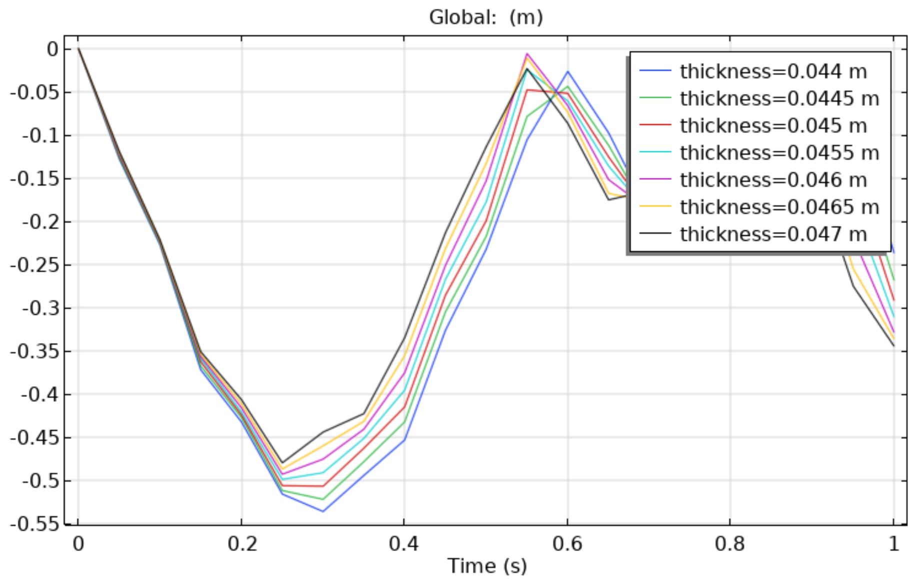

Figure A3: *Deflection-Time plot for parametric sweep of thickness 0.044 to 0.047 meter*

The process is repeated in Figure A4, with ranges becoming smaller and smaller:

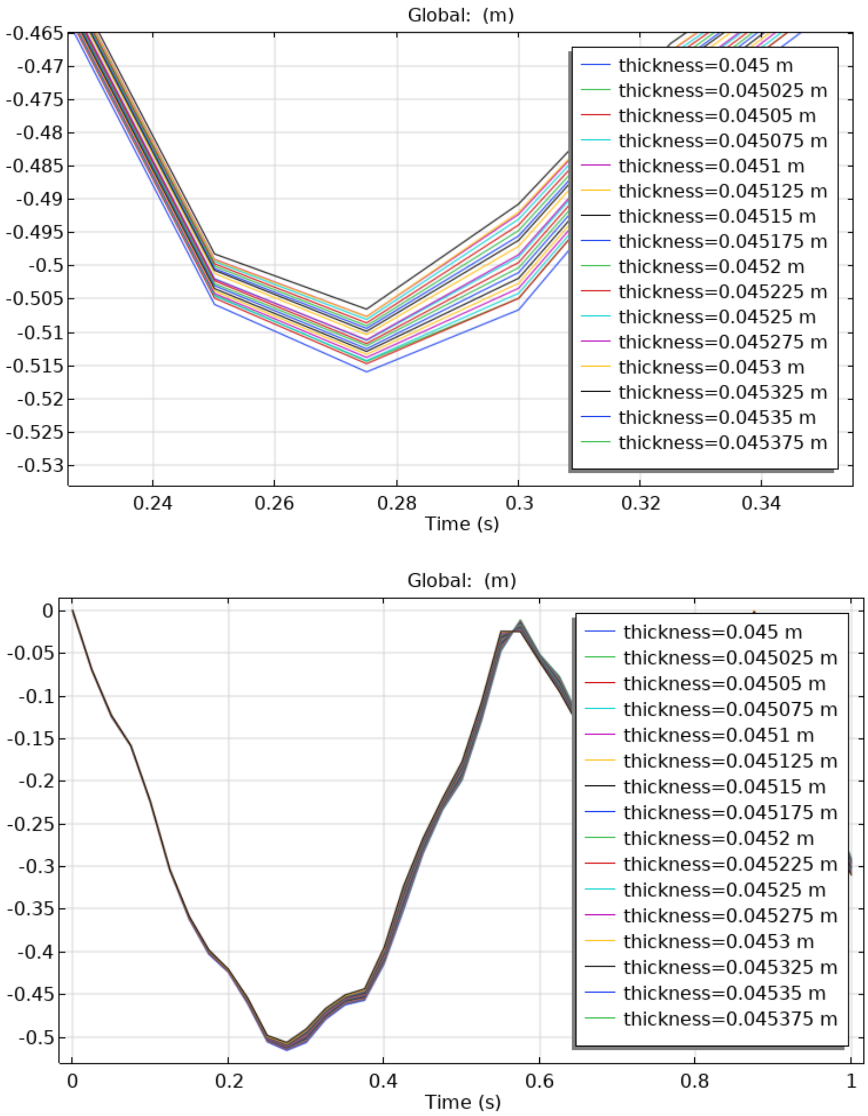

Figure A4: *Deflection-Time plots for smaller range of parametric sweep.*

The output times were refined, giving a less "choppy" graph, and the process was repeated for Figure A5:

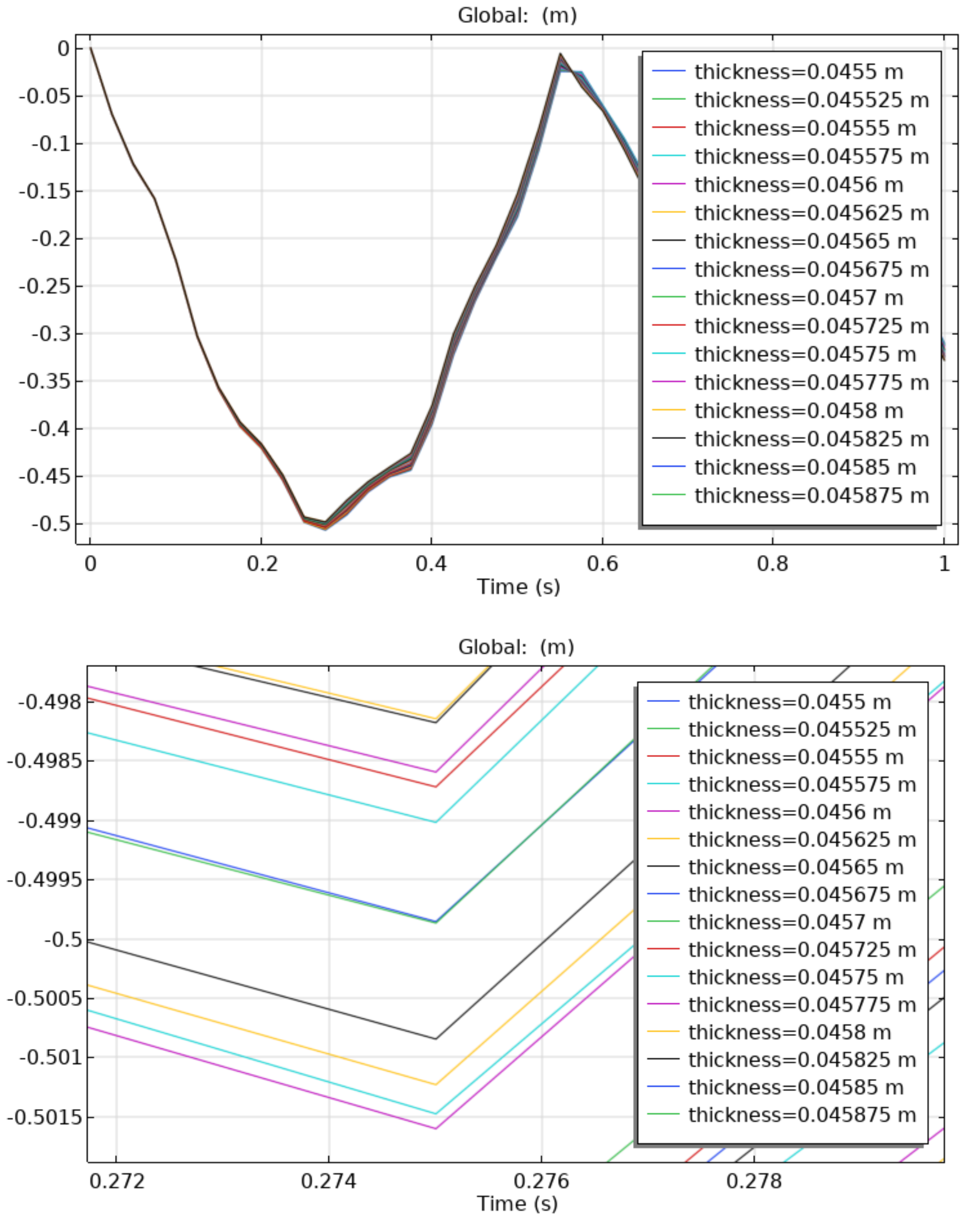

Figure A5: Upper: *Last deflection-time plot for smaller thickness. Lower: zoomed in on values close to maximum permitted end displacement*

By looking near 0.5 end deflection it was concluded that the thickness closest to 0.5 meter maximum end deflection was 0.04585 meter thickness.

9: Appendix B:

Optimization of plywood in Case (c)

First the roller support was added and the simulation ran for 0.0476 meter thickness. As is seen in Figure B1, this gave a maximum displacement larger than permitted. The plywood percent was set to the optimised value in Case (a), i.e. v_w was set to 0.9069.

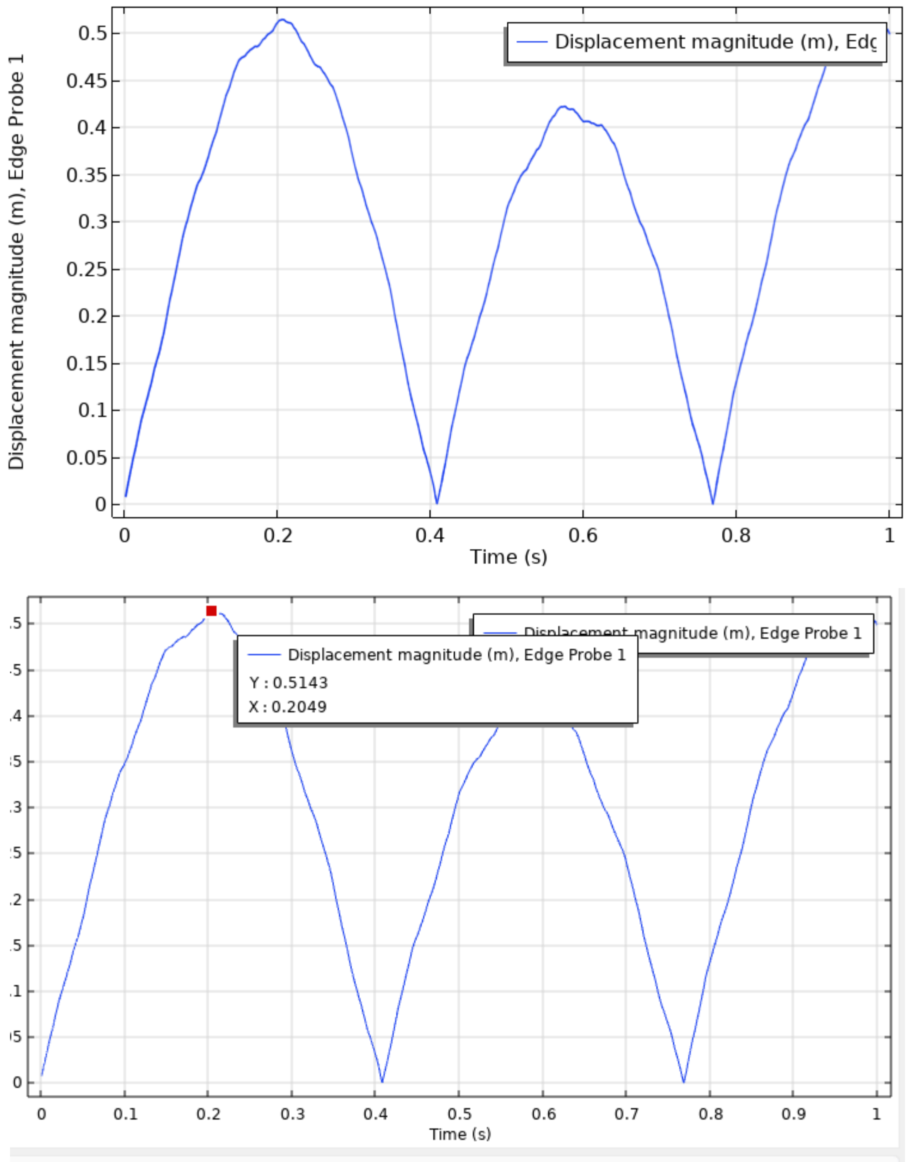

Figure B1: *Displacement magnitude-Time plot for 0.0476 meter thickness. Maximum displacement being around 0.5143.*

To find the optimised value the thickness was decreased to 0.0486 m and the simulation was repeated. This gave a maximum displacement at the plot of 0.4973, which is below the permitted, the optimized value is thus between these two.

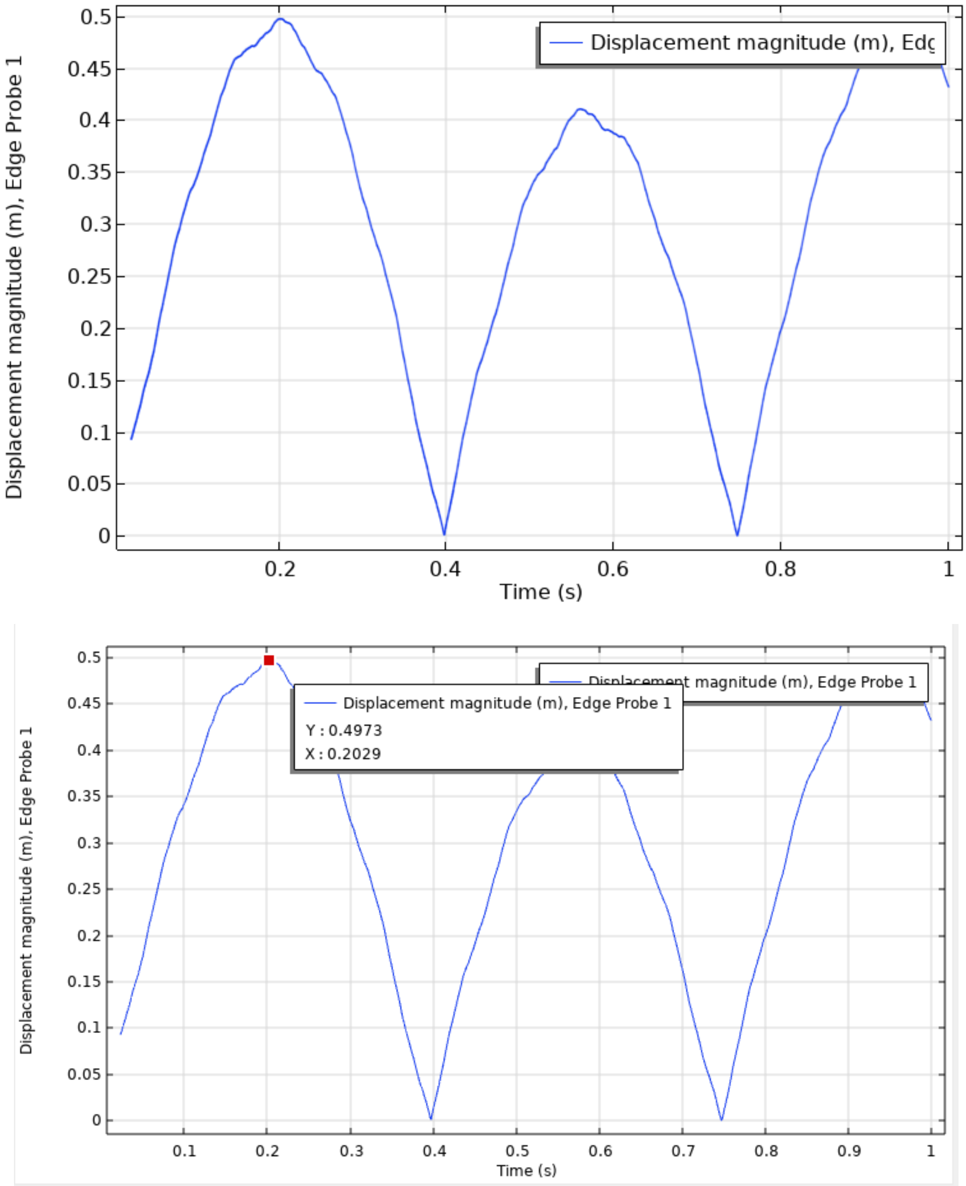

Figure B2: *Displacement magnitude-Time plot for 0.0486 meter thickness.*

A simulation with 0.0481 meter thickness was tried and it gave a maximum displacement larger than 0.5 meter. The above process was repeated until the optimized value was found, this was done by decreasing the simulated time, since only the first cycle is of importance and using a parametric sweep of values between 0.0481 and 0.0486. The optimized value was between 0.0481 and 0.0482 meter thickness.

Case (c) with same thickness as was optimized for Case (b)

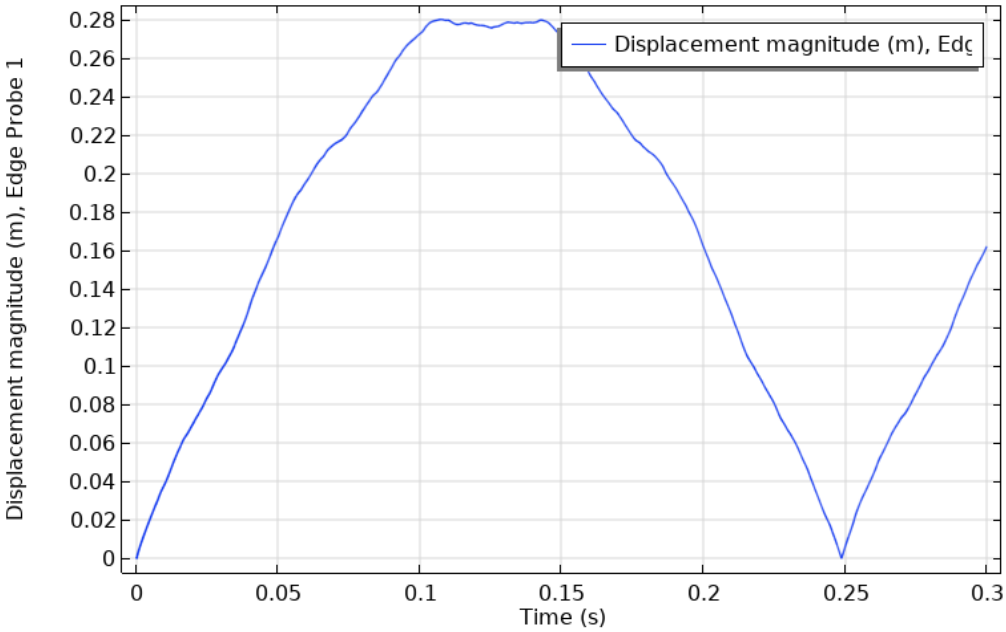

Figure B4: *End displacement for Case (c) for thickness 0.068 meter.*

Optimization of aluminum for Case (c)

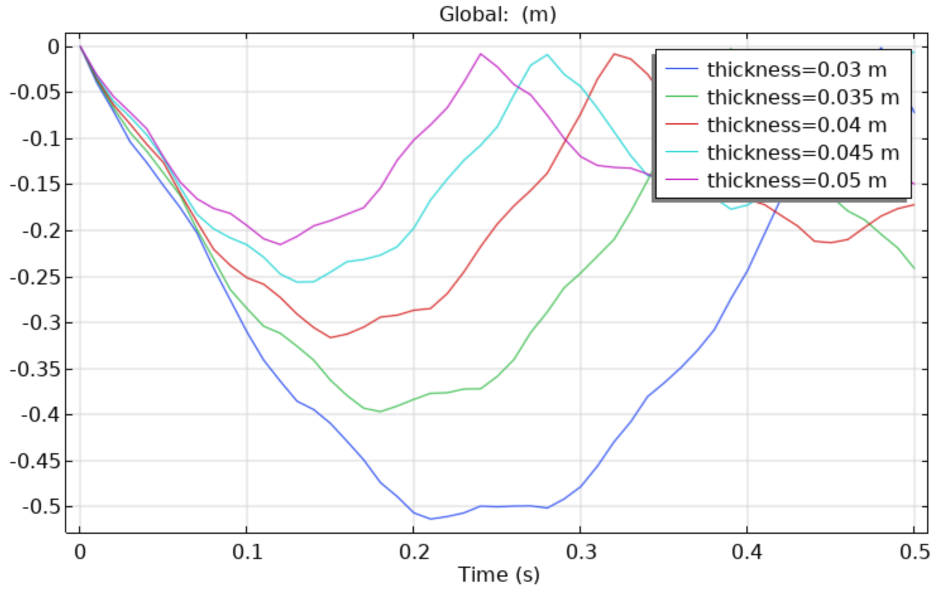

Figure B5: *First rough parametric sweep from 0.03 to 0.05 meter thickness.*

Since 0.03 meter was close to the permitted end displacement 0.5 meter, values between 0.03 and 0.035 were investigated.

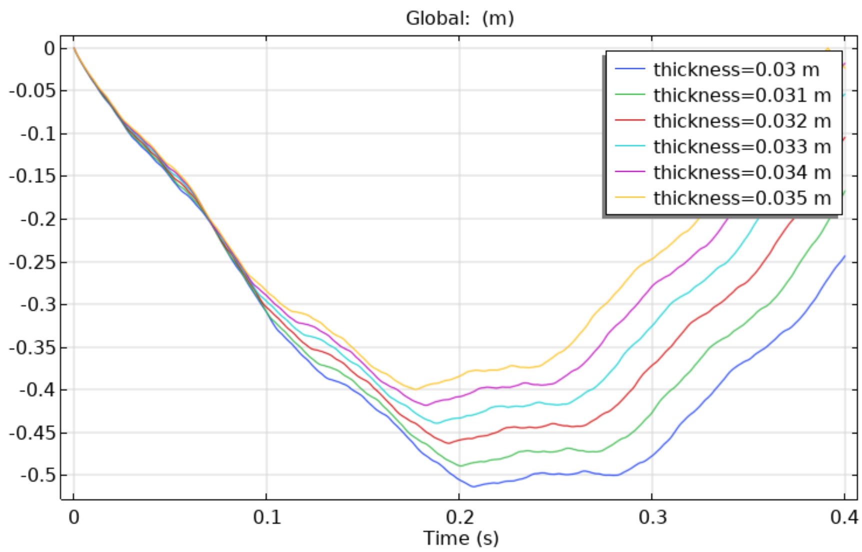

Figure B6: *Finer parametric sweep for 0.03 to 0.035 meter thickness.*

Again the value closest to 0.5 meter end displacement is close to 0.03, a finer sweep is conducted.

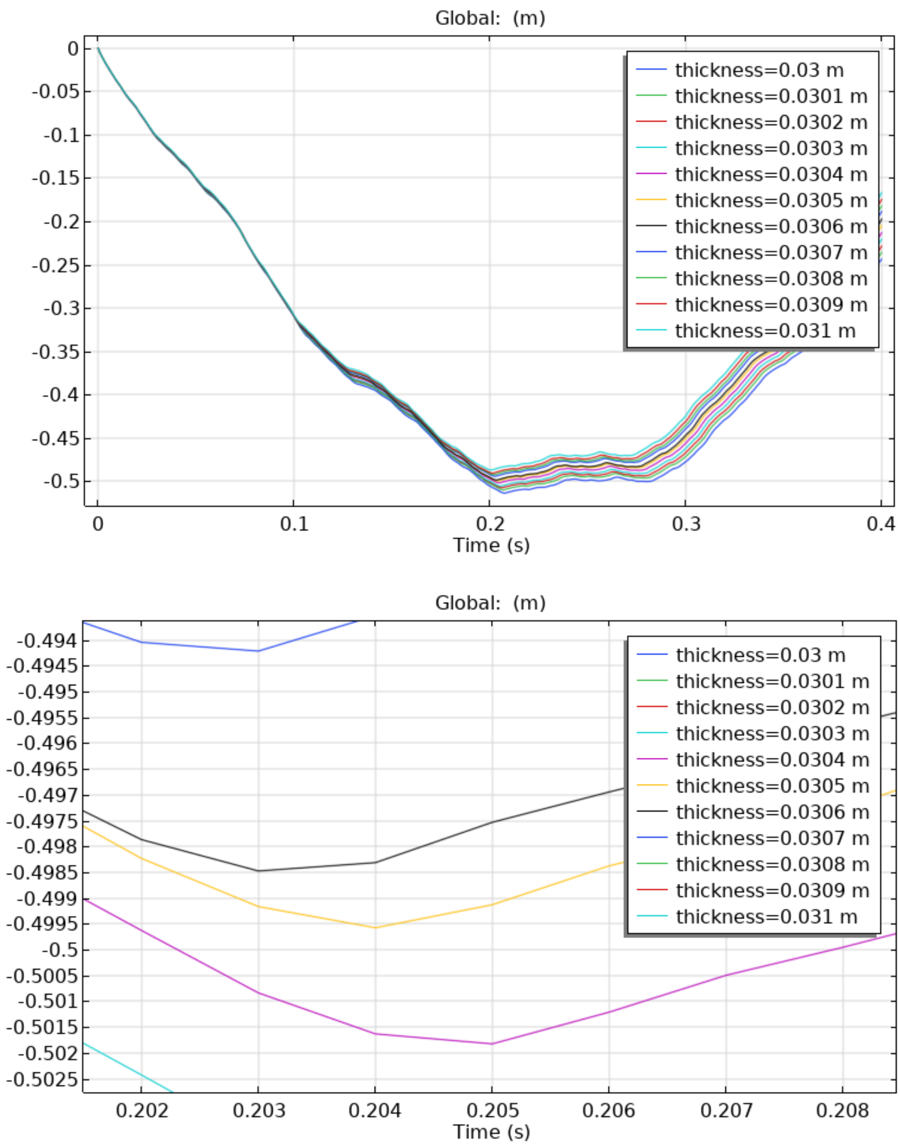

Figure B7: *Upper: Final parametric sweep for the end displacement- time plot for values between 0.03 and 0.035. Lower: zoomed in on values close to 0.5 end displacement.*

An optimized value between 0.0305 and 0.0306 meter thickness is found, the lower of the two is chosen.

10: Appendix C: Geometry Parameters

Table C1: *Geometry Parameters*

Description	Parameter	Value	Unit
Body mass	M_{Body}	70	kg
Body height	H_{Body}	1.80	m
Body width	W_{Body}	0.3	m
Body length	L_{Body}	0.3	m
Body volume	V_{Body}	0.162	m^3
Impact zone	w_F x l_F	0.09	m^2
Feet width	w_F	0.3	m
Feet length	l_F	0.3	m

Values are taken from a rough estimation of a human.

11: Appendix D: Material Parameters

Table D1: *Material Parameters*

Description	Parameter	Value	Unit
Density for Aluminum	ρ_A	2710	kg/m^3
Tensile Yield Stress for Aluminum	$\sigma_{s,A}$	352	MPa
Young's modulus for Aluminum	E_A	68.9	GPa
Poisson's ratio for Aluminum	ν_A	0.33	-
Density for Glassfibre epoxy	ρ_G	1850	kg/m^3
Tensile Yield Stress for glass fibre	$\sigma_{s,G}$	208	MPa
Young's modulus in lateral direction for Glassfibre epoxy	E_L	40	GPa
Young's modulus in transversal direction for Glassfibre epoxy	E_T	8.0	GPa
First Poisson's ratio for Glassfibre epoxy	ν_{LT}	0.25	-
Second Poisson's ratio for Glassfibre-epoxy	ν_{TT}	0.30	-
Shear modulus in for Glassfibre epoxy	G_{LT}	4.0	GPa
Density for Pine	ρ_{Pine}	420	kg/m^3
Tensile Yield Stress for Pine	$\sigma_{s,Ply}$	104	MPa
Young's modulus in grain direction for pine	$E_{x,Pine}$	12	GPa
Young's modulus in radial direction for pine	$E_{y,Pine}$	1	GPa
Young's modulus in circumferential direction for pine	$E_{z,Pine}$	0.7	GPa
First poissons ratio for pine	$\nu_{xy,Pine}$	0.33	-
Second poissons ratio for pine	$\nu_{xz,Pine}$	0.38	-
Third poissons ratio for pine	$\nu_{yz,Pine}$	0.4	-
First shear modulus in for pine	$G_{xy,Pine}$	0.8	GPa
Second shear modulus in for pine	$G_{xz,Pine}$	0.7	GPa
Third shear modulus in for pine	$G_{yz,Pine}$	0.2	GPa

Density for plywood	ρ_{Ply}	703.4	kg/m^3
Tensile Yield Stress for plywood	$\sigma_{s,Ply}$	60	MPa
Young's modulus in grain direction for plywood	$E_{x,Ply}$	9.4	GPa
Young's modulus in y-direction for plywood	$E_{y,Ply}$	6.7	GPa
Young's modulus in z-direction for plywood	$E_{z,Ply}$	1.11	GPa
First poissons ratio for plywood	$\nu_{xy,Ply}$	0.036	-
Second poissons ratio for plywood	$\nu_{xz,Ply}$	0.443	-
Third poissons ratio for plywood	$\nu_{yz,Ply}$	0.427	-
First shear modulus in for plywood	$G_{xy,Ply}$	0.600	GPa
Second shear modulus in for plywood	$G_{xz,Ply}$	0.206	GPa
Third shear modulus in for plywood	$G_{yz,Ply}$	0.186	GPa

Sources: (MatWeb, n.d.), (Wang, 2022), (The Engineering ToolBox, 2004) and (Final Advanced Materials, n.d.)